

Project Development Phase

No. of Functional Features Included in the Solution

Date	02 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

This document lists all the functional features included in the OptiCrop solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

Functional Features Overview

S.No	Feature Name	Feature Description	Module / Component	Status (Done/In Progress/Pending)	Marks Contribution
1	Soil & Climate Input Form	Collects 7 parameters (N, P, K, temperature, humidity, pH, rainfall) via number input fields with validated min/max ranges	Frontend (Streamlit UI)	Done	Core
2	Crop Prediction	Predicts the single best-suited crop using the trained Random Forest model	Backend / ML (app.py + model.pkl)	Done	Core

3	Confidence Score Display	Shows the model's prediction confidence as a percentage	Backend / ML	Done	Core
4	Top-3 Alternate Suggestions	Displays the next two most likely crops as backup options	Backend / ML	Done	Core
5	Input Validation	Enforces sensible min/max ranges on every input field to prevent invalid entries	Frontend (Streamlit UI)	Done	Core
6	Error Handling	Displays friendly error messages if the model file is missing or prediction fails	Backend	Done	Core

Feature Summary

Metric	Count / Value
Total Features Planned	6
Total Features Implemented	6
Core / Must-Have Features	3 (Input Form, Crop Prediction, Confidence Score)
Additional / Nice-to-Have Features	3 (Top-3 Suggestions, Input Validation, Error Handling)
Features Tested & Verified	6 (see Functional and Performance Testing document)

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
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1	User Interface (UI)	2	Input form, results display
2	Backend / Logic	3	Prediction logic, confidence scoring, top-3 ranking
3	Database / Storage	0	Not applicable — stateless app, no database used
4	API / Integration	0	Not applicable — self-contained Streamlit app
5	Security / Authentication	0	Not applicable — public single-page tool, no login required